BTR Agent Skills Toolkit v2.0 — Expanded Edition

> **What's New:** 7 additional skill buckets covering content production, social media automation, bilingual copywriting, analytics/listening, ad platforms, financial optimization, and the complete Human-in-the-Loop approval workflow system.

FULL BUCKET MAP (19 Total Buckets)

#	Bucket	Priority	Status
1	Planning & Strategy	★★★★★	✅ From v1.0
2	Architecture & System Design	★★★★★	✅ From v1.0
3	Frontend & UI/UX Design	★★★★★	✅ From v1.0
4	Full-Stack Development	★★★★★	✅ From v1.0
5	Testing & QA	★★★★☆	✅ From v1.0
6	DevOps, CI/CD & Deployment	★★★★☆	✅ From v1.0
7	Security & Compliance	★★★★☆	✅ From v1.0
8	Git, Code Review & Collaboration	★★★★☆	✅ From v1.0
9	Agent Orchestration & Multi-Agent	★★★★☆	✅ From v1.0
10	Content & Marketing	★★★☆☆	✅ From v1.0
11	Document Processing	★★★☆☆	✅ From v1.0
12	Debugging & Performance	★★★☆☆	✅ From v1.0
13	**🆕 Visual Content Production (Nano Banana Pro)**	**★★★★★**	**NEW**
14	**🆕 Social Media Scheduling & Calendar**	**★★★★★**	**NEW**

#	Bucket	Priority	Status
15	**Bilingual Copywriting (EN/AR)**	★★★★★	**NEW**
16	**Analytics, Listening & Prediction**	★★★★★	**NEW**
17	**Ad Platform Integration & Optimization**	★★★★☆	**NEW**
18	**Financial & API Cost Optimization**	★★★★☆	**NEW**
19	**Human-in-the-Loop Approval & Workflow**	★★★★★	**NEW**

BUCKET 13: Visual Content Production — Nano Banana Pro Engine (★★★★★)

The AI content factory. Images, animations, reels, UGC — all API-driven.

Nano Banana Pro is a professional AI image and video generation platform powered by Google's latest models (including Veo 3 and Runway AI), offering 4K upscaling, smart editing, character consistency, and full commercial rights.[1][2]

Skill	Purpose	Implementation Notes
Nano Banana Pro API Connector	Connect to Nano Banana Pro's API for programmatic image/video generation	API key management, rate limiting, credit tracking[1]
AI Image Generation	Generate social media images, product shots, brand visuals from text prompts	Supports 1K-4K resolution, style transfer, scene generation[3]
Character Consistency Engine	Maintain consistent AI characters/brand ambassadors across all content	Single-prompt character creation, face locking across scenes[4]
UGC Production Pipeline	Create user-generated-content-style videos and images at scale	Character → Scenes → TikTok/Reel-ready exports[4]
AI Video Generation	Create product videos, tutorial animations, social media reels	Access to Veo 3, Runway AI models, 30-second ad creation[2]
Style Transfer & Brand Templating	Apply consistent brand styles across all generated visuals	Color palettes, typography overlay, composition rules
Batch Content Generation	Generate a full month's visual content in one batch run	Template-driven, parameterized generation for 30-day calendars
Image-to-Video Pipeline	Convert static brand images into animated video content	For reels, stories, and short-form video ads[2]
Remotion Video	Programmatic video creation using code	For data-driven videos, personalized

Skill	Purpose	Implementation Notes
Programming	(React-based)	content at scale
Animation & Motion Graphics	Create animated posts, logo reveals, transitions	SVG animation, Lottie, CSS keyframes

API Cost Management for Nano Banana Pro

- Standard resolution (1K): ~6 credits per image[1]
- Higher resolution (2K): ~8 credits per image[1]
- Video generation: varies by duration and model
- **Skill needed:** Credit usage tracker with daily/monthly budget alerts

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BUCKET 14: Social Media Scheduling & Calendar Management (★★★★★)

Content created → Content scheduled → Content published → All automated.

Skill	Purpose	Implementation Notes
Content Calendar Generator	AI generates a full month's content calendar from themes and brand guidelines	Reads brand brief, outputs structured calendar with dates, platforms, copy, visuals[5][6]
Multi-Platform Scheduler	Schedule posts across Instagram, TikTok, LinkedIn, X, Facebook, YouTube	Platform-specific formatting (character limits, hashtags, aspect ratios)[7]
Optimal Timing Engine	Analyze engagement data to determine best posting times per platform	Time zone aware, audience activity patterns, day-of-week optimization[8]
Platform API Connectors	Direct API integration with each social platform for publishing	Facebook Graph API, Instagram API, TikTok API, LinkedIn API, X API, YouTube Data API
Content Adaptation Agent	Auto-adapt one piece of content for each platform's unique format	LinkedIn professional tone → TikTok casual → Instagram visual-first[7]
Calendar Visualization	Display scheduled content in a visual calendar (Notion, Google Calendar, or custom dashboard)	Zapier/Pabbly bridge to calendar tools[5]
Auto-Pause & Reschedule	Pause scheduled content during sensitive events or negative sentiment spikes	Connected to social listening signals[8]

Skill	Purpose	Implementation Notes
Posting Confirmation & Proof	After posting, capture confirmation screenshots/links and log to dashboard	Proof that content was published + scheduled correctly
Zapier/Pabbly Workflow Bridge	Connect generation → scheduling → publishing through automation platforms	No-code automation with 8000+ app integrations[9]

Calendar Visibility Mechanism

The agent must provide a **live calendar view** where you can:

1. See all scheduled posts with thumbnails
2. See which platform each goes to
3. See the exact date/time
4. Confirm via green checkmark that scheduling is complete
5. Click to preview each post as it will appear on-platform

BUCKET 15: Bilingual Copywriting Engine — English & Arabic (★★★★★)

Every piece of content ships in both languages. Not translated — crafted.

Skill	Purpose	Implementation Notes
Bilingual Copy Generator	Generate original copy in both English and Arabic simultaneously	Not machine translation — culturally native writing in each language
Arabic RTL Layout Handler	Ensure all Arabic content renders correctly in right-to-left layouts	CSS direction, font selection (Noto Sans Arabic, Cairo, etc.)
Cultural Tone Calibrator	Adjust messaging for UAE/GCC cultural context	Respect local customs, Islamic values, holiday calendars
Hashtag Research (EN/AR)	Generate trending and niche hashtags in both languages per platform	Arabic hashtag discovery, transliteration handling
Copy Quality Review Agent	Review generated copy for grammar, tone, brand voice consistency	Bilingual proofing, brand terminology dictionary
Ad Copy Variants	Generate A/B/C copy variants for ad testing in both languages	Headlines, descriptions, CTAs — platform-specific character limits
Caption & Hook Writer	Write attention-grabbing hooks for reels/stories in both languages	Follow the BTR ULTIMATE-AI-POWERED-COPYWRITING-FRAMEWORK
Email & Newsletter Copy	Generate email sequences in both languages	Subject lines, body copy, CTA buttons
Landing Page Copy	Generate high-conversion landing page copy (bilingual)	Above-fold hero, benefits, social proof, CTA sections

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BUCKET 16: Analytics, Social Listening & Prediction Engine (★★★★★)

All signals in. Good data in → Smart predictions out.

Skill	Purpose	Implementation Notes
Google Analytics 4 Connector	Pull GA4 data: traffic, conversions, user behavior, events	GA4 Data API, real-time and batch reporting
Social Media Analytics Aggregator	Aggregate engagement metrics from all platforms into one dashboard	Likes, shares, comments, reach, impressions per post
Social Listening Agent	Monitor brand mentions, competitor mentions, industry keywords	Options: Awario API[10], Phyllo API[11], EmbedSocial[12][13], or custom build
Sentiment Analysis Engine	Analyze sentiment of mentions and comments (EN + AR)	Positive/negative/neutral classification with Arabic NLP support
Competitor Intelligence	Track competitor social activity, content performance, ad spend	Benchmarking against industry peers
Trend Detection & Prediction	Identify emerging trends from social signals and search data	Predictive modeling for content planning
Report Generator	Auto-generate weekly/monthly analytics reports with insights	Visual reports with charts, KPIs, recommendations
Attribution Modeling	Connect social/content activity to actual business outcomes	UTM tracking, conversion paths, multi-touch attribution
Custom Social Listening	Build lightweight listening using	Lower cost than enterprise tools,

Skill	Purpose	Implementation Notes
(Build Option)	platform APIs directly	tailored to your specific keywords
Zapier Analytics Bridge	Route analytics data through Zapier to your dashboard/reports	Connect GA4 + social APIs → Notion/Sheets/Dashboard[9]

Recommended Social Listening Stack

- **Budget Option:** Custom build using platform APIs + Zapier routing
- **Mid-Range:** Awario Enterprise (API access, multi-language, mentions tracking)[10]
- **Full-Service:** EmbedSocial (Instagram, TikTok, Facebook, YouTube, X, Threads monitoring with API)[13]
- **Developer-First:** Phyllo API (structured social data, engagement metrics, profile verification)[11]

□ BUCKET 17: Ad Platform Integration & Optimization (★★★★☆)

Ads data flowing in. Budget optimized. Performance maximized.

Skill	Purpose	Implementation Notes
Meta Ads API Connector	Pull/push data from Facebook & Instagram Ads	Campaign performance, audience insights, budget pacing
Google Ads API Connector	Integrate Google Ads data for search/display/YouTube campaigns	Keyword performance, bid optimization, conversion tracking
TikTok Ads API Connector	Connect TikTok advertising data	Creative performance, audience demographics, spend tracking
LinkedIn Ads Connector	B2B ad campaign data integration	Lead gen form data, company targeting performance
Snapchat/X Ads Connectors	Additional platform ad data	As needed for regional campaigns
Cross-Platform Ad Dashboard	Unified view of all ad spend and performance across platforms	Single pane of glass for ROAS, CPC, CPM, CTR
Budget Allocation Optimizer	AI-driven budget rebalancing across platforms and campaigns	Shift budget to best-performing channels automatically
Creative Performance Analyzer	Identify which ad creatives perform best and why	A/B test analysis, creative fatigue detection
Audience Insight Aggregator	Combine audience data from all platforms for unified personas	Cross-platform audience overlaps and gaps

Skill	Purpose	Implementation Notes
Ad Copy/Creative Generator	Generate ad variants using Nano Banana Pro + Bilingual Copywriter	Visual + copy + CTA all generated together

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BUCKET 18: Financial & API Cost Optimization (★★★★☆)

Quality stays high. Waste stays zero. Every API call tracked.

Skill	Purpose	Implementation Notes
API Cost Tracker	Track every API call across all services (LLM, Nano Banana, social APIs) with cost attribution	Real-time cost per agent, per workflow, per output[14]
Budget Alert System	Automated alerts when spending approaches thresholds	Daily/weekly/monthly budget caps with Slack/email notifications[15]
LLM Cost Optimizer	Reduce LLM token waste through prompt optimization, caching, model routing	30-40% of LLM budget is typically wasted on redundant requests[16]
Credit Usage Dashboard	Visual dashboard showing Nano Banana Pro credits, LLM tokens, API calls consumed	Real-time balance and forecast
Cost-Per-Content Calculator	Calculate exact cost of producing each piece of content	Image cost + copy cost + scheduling cost + review overhead
Predictive Cost Forecasting	Forecast next month's costs based on content calendar and historical patterns	Plan budgets before content production starts[15]
Quality-Cost Balancer	Ensure cost cuts don't compromise output quality	Quality scoring system + cost threshold rules
Multi-Tier Model Routing	Route simple tasks to cheaper models, complex tasks to premium models	GPT-4 for strategy, lighter models for formatting tasks[17]

Skill	Purpose	Implementation Notes
Invoice & Billing Reconciliation	Match API invoices to actual content produced	Accountability per campaign, per client, per project
ROI Per Campaign Tracker	Calculate return on investment per marketing campaign	Ad spend + content cost vs. revenue/leads generated

Key Financial Guardrails

- **Daily budget cap** per API service (hard stop, not just alerts)
- **Quality gate:** Never auto-downgrade model quality without human approval
- **Weekly cost review:** Agent generates cost report every Sunday night
- **Per-content cost target:** Set maximum cost-per-post threshold

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□ BUCKET 19: Human-in-the-Loop Approval & Workflow Engine (★★★★★)

The most critical bucket. This is where you stay in control.

The Core Problem You Described

When an agent produces content (designs, posts, videos), you need to:

1. **SEE IT** clearly — where does it appear for review?
2. **GIVE FEEDBACK** — structured, specific, tracked
3. **ITERATE** — go back and forth 2-3 times with versioning
4. **APPROVE** — press a button that locks it as final
5. **KNOW IT SHIPPED** — see proof it was scheduled/published
6. **SEQUENCE IT** — step 1 → step 2 → step 3, with guardrails

Recommended Approval Platform: Miro Board + Trello Hybrid

Why Miro for Review:

- Miro now has native **Workflows for Miro** app with built-in approval steps[18][19]
- Supports single-stage or two-stage approvals
- Complete approval history maintained
- Visual — you can see designs, annotate directly, leave sticky-note feedback
- Apply workflows to specific frames (each post gets its own frame)

Why Trello for Task Sequencing:

- Kanban-style lists: `To Review` → `Feedback Given` → `Revision In Progress` → `Approved` → `Scheduled` → `Published`
- Each card = one content piece with all versions attached
- Checklists on cards for structured feedback
- Due dates for scheduling deadlines
- Zapier connects Miro ↔ Trello for automatic card movement[9]

Complete Approval Workflow Skills

Skill	Purpose	Implementation Notes
Miro Board Auto-Populator	Agent creates a Miro frame for each content piece with visual preview	One frame per post, organized by date and platform
Trello Card Creator	Agent creates Trello cards for each content piece in the pipeline	Card includes: preview image, copy text, platform, scheduled date
Version Control & History	Every revision creates a new version, old versions preserved	v1 → v2 → v3 visible in card attachments or Miro frame layers
Structured Feedback Template	Standardized feedback form: What to change, Why, Priority level	Prevents vague feedback, ensures agent gets clear direction
Feedback Ingestion Agent	Agent reads your structured feedback and applies revisions	Parses feedback → Maps to specific elements → Generates revision
Approval Gate Mechanism	A clear "APPROVE" button/action that locks content as final	Options: Miro approval workflow button[18], Trello card move to "Approved" list, or custom webhook
Approval Notification	You get notified when content is ready for review; agent gets notified when approved	Slack, email, or Trello notification

Skill	Purpose	Implementation Notes
Post-Approval Auto-Scheduler	Once approved, content automatically enters the scheduling queue	Approval triggers → Scheduling agent → Platform API publishing
Scheduling Confirmation Display	After scheduling, you see proof in a calendar view	Google Calendar embed, Notion calendar, or custom dashboard showing: platform + date + time + preview
Sequencing Engine	Enforces strict order: Generate → Review → Feedback → Revise → Approve → Schedule → Publish	No step skips allowed, each gate must be passed
Guardrail Checkpoints	Automated checks at each stage before moving forward	Content guidelines check, brand voice check, prohibited content check, Shariah compliance check
Human-in-the-Loop Pause	Agent pauses and waits for human input at designated checkpoints	Tool-level approval pattern: agent pauses before executing any publishing action[20]
Escalation Rules	If 3 revision rounds don't resolve, escalate to different approach	Auto-escalation after max iterations

The Approval Flow — Step by Step

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┌──────────────────────────────────────────────────────────────────────────────
│                    CONTENT PRODUCTION PIPELINE                              │
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│                                                                             │
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```
│  1. GENERATE ──→ Agent creates content (image/video/copy)│
│     │                                    │
│     ▼                                    │
│  2. PRESENT  ──→ Content appears on Miro board          │
│     │         (one frame per piece, organized by date) │
│     │         + Trello card created in "To Review"     │
│     ▼                                    │
│  3. REVIEW   ──→ You open Miro, see all 30 posts        │
│     │         visually laid out                        │
│     ▼                                    │
│  4. FEEDBACK ──→ You fill structured feedback form      │
│     │         (What to change / Why / Priority)        │
│     │         Trello card moves to "Feedback Given"    │
│     ▼                                    │
│  5. REVISE   ──→ Agent reads feedback, generates v2     │
│     │         Previous version preserved               │
│     │         Trello card moves to "Revision Ready"    │
│     ▼                                    │
│  6. RE-REVIEW ──→ You see v2 alongside v1               │
│     │          (Up to 3 rounds max)                    │
│     ▼                                    │
│  7. APPROVE  ──→ You press APPROVE button in Miro       │
│     │         OR move Trello card to "Approved"        │
│     │         ✅ This locks the content as FINAL       │
│     ▼                                    │
│  8. SCHEDULE ──→ Approved content auto-enters scheduler │
│     │         API connections publish to platforms     │
│     ▼                                    │
│  9. CONFIRM  ──→ Calendar view shows:                   │
│     │         ✅ Platform ✅ Date/Time ✅ Preview      │
│     │         ✅ "Scheduled" status confirmed          │
│     ▼                                    │
│  10. PUBLISH ──→ Content goes live at scheduled time    │
│     │         Post-publish confirmation logged         │
│     ▼                                    │
│  11. ANALYZE ──→ Performance data flows back            │
│               Feeds into next month's planning          │
```

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│                                   │
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How You'll Know Things Happened

Question	Answer
"Where do I see content for review?"	**Miro board** — one frame per post, organized chronologically. Visual + copy side by side.[18]
"How do I give feedback?"	**Structured feedback form** on Trello card or Miro sticky notes: What to change, Why, Priority
"How do I see previous versions?"	**Version attachments** on Trello card (v1, v2, v3) or Miro frame layers
"How do I approve?"	**Press APPROVE** in Miro Workflows[19] OR move Trello card to "Approved" column
"How do I know content was scheduled?"	**Calendar view** (Google Calendar / Notion) shows each post with platform, date, time, thumbnail
"How do I know it was actually published?"	**Confirmation log** — agent captures platform confirmation (post URL, timestamp) and logs to dashboard
"What if I want to stop something?"	**Emergency pause** — move any card to "Hold" list or click "Reject" in Miro to halt the pipeline

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□ INTEGRATION MAP — How All Buckets Connect

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                    ┌──────────────────────┐
                    │ PLANNING (B1)        │
                    │ Strategy & Brief     │
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│ COPY (B15)      │  │ VISUAL(B13)     │  │ DEV (B4)        │
│ EN + AR         │  │ Nano Banana     │  │ SaaS Build      │
└────────┬────────┘  └────────┬────────┘  └────────┬────────┘
         │                   │                   │
         └────────┬──────────┘                    │
                  ▼                               ▼
        ┌──────────────────────┐   ┌──────────────────────┐
        │ APPROVAL (B19)       │   │ DEPLOY (B6)          │
        │ Miro + Trello        │   │ CI/CD Pipeline       │
        └──────────┬───────────┘   └──────────────────────┘
                   │
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        │ SCHEDULE (B14)       │
        │ Social Calendar      │
        └──────────┬───────────┘
                   │
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        ┌──────────────────────┐
        │ PUBLISH              │
        │ Platform APIs        │
        └──────────┬───────────┘
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         ▼         ▼         ▼
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│ANALYTICS       │ │ ADS        │ │ FINANCE        │
│(B16)           │ │ (B17)      │ │ (B18)          │
│GA4+Listen      │ │All Plat    │ │Cost Track      │
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    │ PREDICTION   │
    │ Next Month   │
    │ Planning     │
    └──────────────┘
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⚡ UPDATED PRIORITY LOADING ORDER

1. **LOAD FIRST (Foundation):** Planning → Architecture → Frontend/UI → Full-Stack → **Approval Workflow**
2. **LOAD SECOND (Content Engine): Visual Production** → **Bilingual Copy** → **Social Calendar** → Content Marketing
3. **LOAD THIRD (Quality & Velocity):** Testing → Security → Git/Code Review → DevOps/Deployment
4. **LOAD FOURTH (Intelligence): Analytics/Listening** → **Ad Platforms** → **Financial Optimization**
5. **LOAD ON-DEMAND:** Document Processing → Debugging → Agent Orchestration (for meta-building)

UPDATED QUICK-START PACKS

SaaS Builder Pack

- Writing Plans + Executing Plans + Brainstorming
- Next.js + React + Tailwind + Stripe + Clerk Auth
- UI Design System + SaaS Pricing Page
- Vercel Deploy + GitHub Actions
- TDD + Systematic Debugging

Content Production Pack (NEW)

- Nano Banana Pro API Connector + Batch Content Generation
- Bilingual Copy Generator + Cultural Tone Calibrator
- Content Calendar Generator + Multi-Platform Scheduler
- Miro Board Auto-Populator + Trello Card Creator
- Approval Gate Mechanism + Post-Approval Auto-Scheduler

Analytics & Optimization Pack (NEW)

- GA4 Connector + Social Media Analytics Aggregator
- Social Listening Agent (Awario/Phyllo/EmbedSocial)
- Cross-Platform Ad Dashboard + Budget Allocation Optimizer
- API Cost Tracker + Predictive Cost Forecasting
- Report Generator + ROI Per Campaign Tracker

Shariah Compliance Guardrails Pack (CUSTOM)

- Content screening for prohibited imagery/messaging
- Financial transaction compliance (no riba elements)
- Cultural sensitivity checker for GCC market
- Arabic language quality validator
- Brand voice consistency across both languages